

VANCE AFB, OK, USA

WMO: 723535

Lat: 36.333N

Lon: 97.917W

Elev: 1306

StdP: 14.02

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13909

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	15.6	0.6	5.9	14.2	5.6	7.7	20.5	33.2	38.6	28.9	43.3	10.9	10	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	101.7	73.4	99.1	73.6	96.2	74.1	78.3	90.3	77.2	89.7	76.2	88.8	13.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	139.9	81.6	74.2	134.0	81.1	72.9	128.4	80.7	42.7	90.7	41.5	89.5	40.5	88.7	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.5	26.0	23.6	4.4	106.2	4.9	2.5	0.9	107.9	-1.9	109.4	-4.6	110.8	-8.1	112.6
			4.1	80.3	4.6	1.5	0.8	81.3	-1.9	82.2	-4.5	83.1	-7.8	84.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	35.9	41.8	49.8	57.4	68.4	78.6	83.0	81.6	73.6	61.0	48.4	37.4
	DBStd	18.64	10.26	11.85	10.93	8.92	7.79	6.00	5.08	5.46	7.56	9.13	9.26	8.79
	HDD50	1455	447	277	140	33	0	0	0	0	0	19	138	401
	HDD65	3908	902	651	481	256	56	1	0	0	19	184	501	857
	CDD50	5036	9	47	132	254	570	857	1022	979	708	359	89	10
	CDD65	2017	0	1	9	27	161	408	557	514	278	59	3	0
	CDH74	22941	1	21	109	357	1586	4629	6955	6039	2642	570	32	0
	CDH80	11676	0	4	25	81	600	2393	3937	3291	1177	165	3	0
(14) Wind	WSAvg	11.8	11.7	12.1	13.4	13.6	11.7	12.3	10.7	10.3	10.6	11.9	11.8	11.3
(15) Precipitation	PrecAvg	32.1	1.0	1.2	2.3	3.3	3.9	4.7	3.0	3.6	2.6	2.7	2.0	1.5
	PrecMax	49.5	2.7	4.1	7.1	7.7	9.9	13.0	6.0	8.9	5.3	7.4	5.7	3.8
	PrecMin	19.8	0.0	0.0	0.0	0.4	0.4	0.7	0.1	0.2	0.0	0.1	0.0	0.1
	PrecStd	7.1	0.8	1.1	1.7	1.8	2.4	2.9	1.4	2.1	1.3	1.9	1.6	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.4	76.7	83.8	86.7	95.1	101.8	105.6	105.3	99.5	89.7	78.6	67.3
		MCWB	52.8	58.8	61.8	64.8	72.4	74.2	73.0	73.0	71.7	70.0	62.5	53.6
	2%	DB	62.9	71.0	76.7	82.2	90.3	98.1	102.1	101.3	94.2	84.5	72.7	62.2
		MCWB	49.7	54.5	60.8	63.6	71.9	74.0	73.4	73.6	67.8	58.9	52.4	42.2
	5%	DB	57.4	65.9	72.0	78.3	86.7	95.2	99.3	98.8	90.8	80.6	68.1	56.2
		MCWB	47.6	53.9	58.6	62.2	70.6	74.2	74.2	73.1	72.9	65.2	57.1	48.0
	10%	DB	52.3	60.7	67.3	74.0	82.8	92.1	96.2	95.2	87.3	76.5	63.5	52.3
		MCWB	44.3	51.0	56.1	60.5	69.1	73.4	74.6	74.0	71.9	63.6	54.7	44.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.8	63.0	66.4	69.9	77.0	79.5	80.1	79.7	78.0	74.7	65.4	62.4
		MCDB	63.4	67.4	75.7	79.7	85.9	92.2	90.5	91.0	87.1	83.0	72.5	65.1
	2%	WB	52.1	59.1	63.9	67.9	74.8	77.7	78.1	77.8	76.4	72.0	62.1	53.8
		MCDB	58.9	67.0	71.8	76.9	84.6	91.4	90.5	90.0	86.4	78.8	68.5	57.5
	5%	WB	48.1	55.9	61.3	65.8	73.0	76.4	77.0	76.6	74.8	69.2	59.5	49.1
		MCDB	56.5	63.2	67.9	73.7	82.9	89.7	90.1	89.1	84.8	75.3	65.5	55.8
	10%	WB	44.7	51.5	58.3	62.9	71.2	75.1	76.0	75.4	73.4	66.2	56.5	45.2
		MCDB	51.7	61.6	66.1	71.1	80.4	87.5	89.3	88.1	83.2	74.0	62.7	51.5
(35) Mean Daily Temperature Range	5% DB	MDBR	22.0	23.2	23.3	23.7	21.8	22.1	22.6	21.9	20.8	22.2	22.4	20.3
		MCDBR	30.7	31.1	29.9	29.9	25.9	25.5	26.7	27.1	24.7	26.6	28.0	28.0
	5% WB	MCWBR	18.7	18.3	16.8	15.9	11.1	8.1	6.8	7.4	8.5	12.6	16.6	18.8
		MCWBR	27.3	25.9	24.5	24.0	22.4	22.9	22.4	21.2	21.0	19.9	23.2	25.1
(40) Clear-Sky Solar Irradiance	taub		0.286	0.294	0.335	0.373	0.406	0.417	0.420	0.424	0.380	0.343	0.301	0.282
		taud	2.501	2.496	2.402	2.319	2.304	2.311	2.329	2.265	2.367	2.456	2.507	2.522
	Ebn at Noon		291	302	296	288	279	274	273	270	277	279	283	286
		Edh at Noon	25	29	35	40	41	41	40	42	36	30	25	23
(44) All-Sky Solar Radiation	RadAvg		867	1078	1413	1755	1940	2192	2176	1961	1635	1227	922	735
	RadStd		84	156	98	123	193	174	121	112	119	153	115	74

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page